

Forecasting the Winner of a Live Tennis Match

Charles Xie^{1‡}

Aneesh Muppidi²

¹Natick High School, Natick, Massachusetts, USA charlesxie157@gmail.com

²University of Oxford, Oxford, United Kingdom aneesh.muppidi@magd.ox.ac.uk

Abstract

Live tennis forecasting requires models to adapt continually as the score and in-match performance change. Although existing models capture either the structural progression of a match or patterns in live data, fewer approaches integrate both sources of information into a single model. This study investigates whether combining hierarchical Markov recursion, Bayesian live serve updates, and gradient boosting - in addition to pre-match features like an elo system - can improve match-win probability estimation.

The analysis uses 8,222 Grand Slam matches containing 1,505,355 point-level prediction states. Five models were evaluated using a chronological split, with matches from 2011–2021 used for training, 2022 for validation, and 2023–2024 for testing. The stacked Markov ensemble achieved accuracies of 76.06%, 82.15%, and 88.34% at 25%, 50%, and 75% match progress, respectively, while producing corresponding log losses of 0.4753, 0.3530, and 0.2002.

These findings demonstrate that combining explicit score structure with data-driven adjustments can improve both winner classification and probability quality, providing an strong foundation for live tennis forecasting.

 [Code](#)

1 Introduction

In recent decades, advances in real-time data collection and digital betting platforms have contributed to the rapid growth of live sports betting (Ding, 2025). Consequently, live betting has developed into a substantial market. Tennis accounts for a significant part of this market because odds can be updated after every point. During the 2008 Wimbledon final between Roger Federer and Rafael Nadal, for example, approximately £50 million in wagers were matched on Betfair alone (Huang, 2011).

Unlike pre-match forecasting, in live tennis prediction, an influx of live statistics is provided. This can be seen with the growing use of technologies such as Hawk-Eye, substantially increasing the amount of shot-level and contextual information that can be collected throughout a tennis match (Fitzpatrick et al., 2024). Although this information may contain useful predictive signals, they can also be harmful if not utilized correctly. For instance, a too little or too much live data can trick a model into thinking in a false direction (Gollub, 2021). Thus, a successful live model must determine a way to balance the incorporation of live data (Ciobanu, 2023).

Despite potential challenges, tennis is particularly well suited to live-match modeling because of its discrete, hierarchical scoring structure and variety of data. For tennis scoring, points determine

[‡]Correspondence to: charlesxie157@gmail.com. Version of July 27, 2026.

games, games determine sets, and sets determine matches, allowing win probability to be calculated across each level of the scoring system (O'Malley, 2008). The variety of live tennis data have also supported models that analyze player behavior and predict shot outcomes using spatiotemporal match information (Wei et al., 2013). This study combines these two ideas: a hierarchical Markov recursion with pre-match Elo ratings, Bayesian updating of live serving performance, and a gradient-boosted ensemble to examine whether structural and data-driven methods can jointly improve live win-probability estimation.

2 Related Work

Live tennis prediction has received meaningful research attention, although it remains less developed than pre-match forecasting. Klaassen & Magnus (2003) introduced a hierarchical Markov model that updates match-winning probability from the current score, server, and fixed player-specific serve probabilities. Because these serve probabilities remain unchanged during the match, the model does not incorporate evidence from a player's live performance. Gollub (2017) extended hierarchical Markov prediction by deriving serve probabilities from Elo-based pre-match forecasts and updating them using a beta-binomial model during the match. For live matches, this model reached a maximum accuracy of 76.5% after averaging all points.

Lerner et al. (2019) built on Gollub (2017)'s model by introducing DeepTennis, an LSTM model designed to estimate the probabilities of winning a match after each point of a tennis match. Their model combined sequential point-by-point match data with pre-match probability estimates, achieving an overall test accuracy of 79.5%. Among the live tennis studies reviewed, this was the highest reported overall accuracy; however, because it was calculated under a different point-level evaluation framework, it serves as a contextual benchmark rather than a directly comparable baseline for this study.

Although this study focuses primarily on live-match prediction, pre-match models remain valuable because they provide an initial estimate of player strength before any in-match evidence becomes available. One prominent model is the creation of an Elo system for tennis, which has been explored in-depth by multiple papers already. Vaughan Williams et al. (2021) found that Elo-based methods generally provided stronger forecasts of current player performance than official rankings, supporting their use as pre-match probability estimates. More concretely, (Kovalchik, 2016) compared eleven pre-match tennis forecasting approaches and found that the career-history Elo model performed second best, right under the market consensus odds, achieving 70% accuracy overall. This further corroborates the use of Elo as a reliable pre-match prior.

3 Data

This project uses public tennis datasets, consisting of Grand Slam point-by-point data and ATP/WTa tour-level match results (Sackmann, 2024a;b;c). The Grand Slam point-by-point data provides the live match information used to construct prediction states, including the server, point winner, set number, game score, point score, rally length, serve speed, break-point indicators, aces, double faults, and first and second serve outcomes. The ATP and WTA match results are used separately to build pre-match Elo ratings for each player.

3.1 Data Preprocessing

Following data cleaning, the prepared dataset contained 8,222 matches and 1,505,355 point-level prediction states. Table 1 summarizes the composition of the sample, its chronological division into training, validation, and test periods, and the proportion of matches successfully linked to pre-match Elo ratings. Point-state counts are reported for the prepared dataset before the Elo merge.

Table 1: Composition, filtering, and Elo coverage of the prepared dataset.

Quantity	Matches	Point states
<i>Prepared sample</i>		
Total	8,222	1,505,355
ATP	4,181	–
WTA	4,041	–
<i>Temporal split</i>		
Training, 2011–2021	6,774	1,238,084
Validation, 2022	473	87,439
Testing, 2023–2024	975	179,832
<i>Filtering and coverage</i>		
Raw matches excluded (21.8%)	2,286	–
Matched to Elo ratings (96.0%)	7,896	–

A match was retained only if its point sequence produced a valid, non-tied final set score, the winner had completed either two or three sets, more than 20 points were recorded, and the server and point winner were identified as either player 1 or player 2. These criteria removed walkovers, unscored matches, and records whose outcomes could not be resolved reliably. Retirements were not excluded solely on the basis of their match label; eight retirement-labeled matches satisfied the remaining criteria and were retained.

After filtering, the point-by-point records were reorganized so that each row represents the match state immediately before a point was played. Each state contains the current set, game, and point scores, the serving player, the match format, and other information available at that stage of play. The eventual match winner is stored separately as the prediction target, indicating whether player 1 won. This organization allows the models to estimate player 1’s match-win probability using only information available at the corresponding point in the match.

The retained match states were then converted into the numerical format required by the models. The most important transformation involved the tennis point score, since the Markov recursion operates on numerical game states rather than textual labels such as 15, 30, 40, and advantage. Table 2 presents the encoding used throughout the analysis.

Table 2: Numerical encoding of tennis point scores.

Raw point score	Encoded value
0	0
15	1
30	2
40	3
AD	4

This encoding preserves the ordering of points within a game and enables the score state to be passed directly into the recursive model. The remaining match-state variables were standardized in a comparable manner. Completed set winners were accumulated to track the number of sets won by each player, server identity was converted into an indicator for whether player 1 was serving, and games tied at 6–6 were designated as tiebreak states.

Once the structural match state had been standardized, raw point-level events were converted into running features that summarize performance earlier in the match. Individual events, such as a single ace, serve, or point outcome, may be noisy in isolation, whereas cumulative events like serve rates and game/set outcomes provide the models with a more stable representation of current form. Each

feature is calculated using only points completed before the prediction state, preventing lookahead leakage. Table 3 summarizes the principal variable groups in the processed dataset.

Table 3: Principal variable groups in the processed point-level dataset.

Feature group	Examples	Description
Score state	<code>best_of</code> , <code>p1_sets</code> , <code>p2_sets</code> , <code>p1_games</code> , <code>p2_games</code> , <code>p1_score</code> , <code>p2_score</code> , <code>p1_serving</code> , <code>tiebreak</code>	Describes the match format and the current set, game, point-score, and server state.
Match outcome	<code>winner</code> , <code>y</code>	Stores the final match result used exclusively as the prediction target.
Live serve performance	<code>p1_serve_rate</code> , <code>p2_serve_rate</code> , <code>p1_serve_n</code> , <code>p2_serve_n</code>	Tracks each player’s observed serve-point win rate and corresponding sample size before the current point.
Rally and ace context	<code>rally_avg</code> , <code>recent_rally_avg</code> , <code>p1_ace_rate</code> , <code>p2_ace_rate</code>	Summarizes rally length and ace performance from previously completed points when those measurements are available.
Derived state summaries	<code>sets_diff</code> , <code>games_diff</code> , <code>score_diff</code> , <code>pts_played</code>	Provides compact representations of the evolving match state from player 1’s perspective.

3.2 Elo

Pre-match player strength was represented using chronological Elo ratings constructed from historical ATP and WTA match results. Following the career-adjusted Elo formulation described by Kovalchik (2016), all players were initialized at 1500, and ratings were updated chronologically using a player-specific update factor that declined with career match experience. For a match between players i and j , player i ’s expected win probability was

$$q_{i,t} = \frac{1}{1 + 10^{(R_{j,t} - R_{i,t})/400}}, \quad (1)$$

where $R_{i,t}$ and $R_{j,t}$ are the players’ ratings immediately before the match. After the outcome was observed, the rating was updated as

$$R_{i,t+1} = R_{i,t} + K_{i,t} (S_{i,t} - q_{i,t}), \quad K_{i,t} = \frac{250}{(m_{i,t} + 5)^{0.4}}, \quad (2)$$

where $S_{i,t} \in \{0, 1\}$ denotes the match outcome and $m_{i,t}$ is the number of matches previously completed by player i . The declining update factor allows ratings to adjust rapidly for inexperienced players while becoming more stable as additional match history accumulates.

Ratings were recorded before each match result was incorporated, ensuring that the Elo features contained only information available before the match. They were then linked to the Grand Slam point-by-point records using standardized player names, tournament information, and match metadata.

3.3 Evaluation Split

The evaluation uses a time-based split. Matches from 2011 to 2021 are used for training, 2022 is used for validation and parameter tuning, and matches from 2023 onward are used for testing. This setup makes the evaluation more realistic because player strength and playing conditions change over time. Therefore, testing later matches better reflects the actual forecasting problem, where a model trained on past data must make predictions about future matches.

4 Methods

To evaluate how different information sources and modeling assumptions affect live prediction, the study compares five models: a symmetric Markov anchor, an Elo-asymmetric Markov model, a serve-shrink Markov model, a histogram gradient boosting model, and a stacked Markov ensemble that combines structural Markov predictions with machine-learning adjustments. Each model serves a different purpose and isolates the contribution of a particular information source or modeling approach.

4.1 Markov Score Recursion

The backbone of every model (excluding the baseline HGBM) is a markov recursion model.

This model converts point-level serve probabilities into a match-win probability using the scoring rules of tennis. Let p_1 be player 1’s probability of winning a point on serve and let p_2 be player 2’s probability of winning a point on serve. Player 1’s probability of winning the next point is

$$q = \begin{cases} p_1, & \text{when player 1 serves,} \\ 1 - p_2, & \text{when player 2 serves.} \end{cases} \quad (3)$$

The recursion proceeds hierarchically from points to games, games to sets, and sets to the match. At each level, the probability of winning the current state is computed from the two possible next states. For example, if $G(i, j)$ is player 1’s probability of winning a game from point score (i, j) , then

$$G(i, j) = qG(i + 1, j) + (1 - q)G(i, j + 1). \quad (4)$$

Terminal states follow the official scoring rules, including two-point margins, server alternation, and tiebreaks. This keeps the tennis structure fixed and reduces the statistical problem to estimating p_1 and p_2 accurately.

4.2 Symmetric Markov Baseline

The symmetric Markov baseline treats both players as equally strong. It uses one global serve-point win probability:

$$p_a = p_b = p$$

The value of p is selected on the validation season. Because both players receive the same serve probability, this model is purely score-based. If two matches have the same score state, the symmetric model gives them the same win probability regardless of player identity.

This baseline was created to measure how much information tennis score can capture alone. It is also the simplest structural anchor for the rest of the project.

4.3 Elo-asymmetric Markov model

The Elo-asymmetric model keeps the same score recursion but allows the two players to start out with different serve probabilities. The model uses the pre-match Elo difference:

$$d = \text{Elo}_1 - \text{Elo}_2$$

Then, the Elo gap is converted into a serve-probability edge:

$$e = \text{clip}(\alpha d, -0.15, 0.15)$$

where α is a validation-tuned slope. The serve probabilities are then

$$p_a = \text{clip}(\beta + e, 0.45, 0.88),$$

$$p_b = \text{clip}(\beta - e, 0.45, 0.88)$$

where β is the tour-average baseline serve probability, and the clipping bounds prevent extreme Elo differences from producing unrealistic serve probabilities. Overall, this model lets stronger players begin a match with a higher implied chance of winning serve points, which allows the model to start away from 50-50 when the players differ in strength.

4.4 Serve-shrink Markov model

The serve-shrink model extends the Elo-asymmetric model by incorporating each player’s serving performance from earlier in the current match. It updates the pre-match serve probabilities when live results indicate that a player may be performing differently from the initial expectation.

Live serve-point win rates can be unstable early in a match because they are based on small samples. For example, a player who wins 9 of their first 10 serve points has a live serve-win rate of 90%, even if their historical serve rate is only 70%. Such an extreme early rate may reflect random variation rather than a lasting change in performance. Hence, following a simplified version of [Colin & Bechler \(2015\)](#)’s Bayesian shrinkage formula, the model combines the live rate with the pre-match probability to give a better estimate of player serve rate. Bayesian shrinkage follows the general equation

$$\hat{\theta} = \frac{n}{n + \kappa} \hat{\theta}_{\text{live}} + \frac{\kappa}{n + \kappa} \theta_{\text{prior}}, \quad (5)$$

where $\hat{\theta}_{\text{live}}$ is the live estimate, θ_{prior} is the pre-match estimate, $\hat{\theta}$ is the resulting updated estimate, n is the number of observed serve points, and κ is a prior pseudo-count controlling the amount of shrinkage.

Let π_1 and π_2 denote the Elo-based prior serve-point win probabilities for players 1 and 2, respectively. Let r_1 and r_2 denote their observed in-match serve-point win rates, and let n_1 and n_2 denote the corresponding numbers of serve points observed before the current match state. The updated serve probabilities are

$$p_a = \frac{n_1 r_1 + \kappa \pi_1}{n_1 + \kappa}, \quad p_b = \frac{n_2 r_2 + \kappa \pi_2}{n_2 + \kappa}. \quad (6)$$

The live rate receives weight $n_i/(n_i + \kappa)$, while the prior receives weight $\kappa/(n_i + \kappa)$. When no serve points have been observed, the updated probability equals the prior. As the number of observed serve points increases, the live rate receives greater weight.

The parameter κ determines how quickly the model adapts to live performance. A larger value keeps the estimate closer to the pre-match prior for longer, while a smaller value makes the model respond more quickly to the current match. Its value is selected using validation log loss.

Finally, p_a and p_b are passed into the Markov score recursion together with the current score state. The recursion then converts the updated serve-point probabilities into the probability that player 1 wins the match.

4.5 Histogram gradient-boosting baseline

The histogram gradient-boosting model serves as a non-structural machine-learning baseline and was implemented using scikit-learn’s `HistGradientBoostingClassifier` (Pedregosa et al., 2011). Unlike the Markov models, it does not explicitly encode the hierarchical scoring rules of tennis or use probabilities generated by the Markov recursion. Instead, it learns the probability that player 1 wins the match directly from the current score state and constructed live-match features.

Let z represent the score and live-performance features available at the current match state. The baseline prediction is

$$\hat{p}_H = g_\phi(z), \quad (7)$$

where g_ϕ is a histogram gradient-boosting classifier and $\hat{p}_H$ is the predicted probability that player 1 wins the match.

This baseline tests whether a flexible machine-learning model can learn live win probability directly from the available features without using the structural assumptions of the Markov recursion. Its hyperparameters are selected using validation log loss, and the final model is refit using the combined training and validation data before evaluation on the held-out test set.

XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) were also evaluated using the same training and validation procedure. Because the histogram gradient-boosting classifier achieved the lowest validation log loss, it was selected as the machine-learning baseline.

4.6 Stacked Markov Ensemble

The final model for this paper is the Stacked Markov Ensemble. The stacked Markov ensemble combines predictions from the Elo-asymmetric and serve-shrink Markov models with score-state and live-match features. These inputs are passed to a Histogram Gradient Boosting Classifier (HGBC), which learns nonlinear adjustments and produces the final match-win probability. This approach preserves the structures of the other two models while adding a machine learning layer to tune the outputs.

The model pipeline is shown in Figure 1.

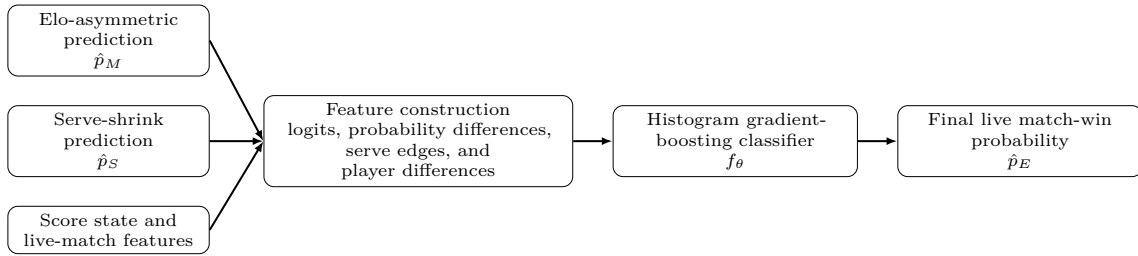


Figure 1: Stacked Markov ensemble pipeline. Structural predictions and live-match information are transformed into features and combined by a Histogram Gradient Boosting Classifier.

Let $\hat{p}_M$ denote the Elo-asymmetric Markov probability, $\hat{p}_S$ denote the serve-shrink Markov probability, and z denote the additional model features. The final ensemble prediction is

$$\hat{p}_E = f_\theta(\hat{p}_M, \hat{p}_S, z). \quad (8)$$

The function f_θ is a Histogram Gradient Boosting Classifier. This allows the ensemble to learn nonlinear relationships, such as whether a live serve rate is reliable given its sample size or whether current performance differs from the pre-match expectation.

Model hyperparameters were selected using validation log loss. The final model is then refit using the combined training and validation data before evaluation on the held-out test set.

5 Experimental Results

The models are evaluated on the test set, which contains matches strictly from 2023 and 2024. The accuracies of the models were collected at fixed fractions of match progress. For example, at 25% match progress, the model receives the score state and live features from roughly one quarter of the way through each match. The same idea is used at 50% and 75%. This setup measures how performance changes as more of the match becomes available. Additionally, the models’ accuracies over all points were also examined.

The project reports two metrics: log loss and accuracy. These metrics will be explained in-depth in the following sections.

Hyperparameters are selected and tuned using the validation season. Final performance is evaluated once on the held-out test period.

5.1 Accuracy

Accuracy is the most widely used and intuitive measure of model performance. It represents the proportion of predictions in which the model correctly identifies the eventual match winner, making it useful for straightforward comparisons across models.

Table 4: Model accuracy at fixed match-progress checkpoints and across all points

Model	25%	50%	75%	All Points
Symmetric Markov	0.6851	0.7703	0.8544	73.21%
Elo-asymmetric Markov	0.7575	0.8050	0.8648	77.56%
Serve-shrink Markov	0.7564	0.7946	0.8720	77.68%
HGBM	0.6944	0.7918	0.8738	73.98%
Markov Ensemble	0.7606	0.8215	0.8834	77.84%

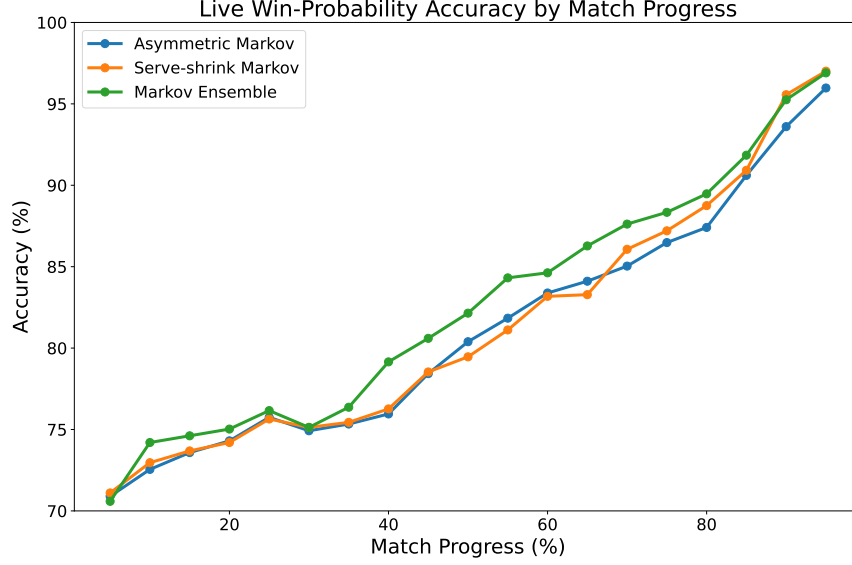


Figure 2: Accuracies of the top three performing models from 5% to 95% match progress.

The stacked Markov ensemble achieved the highest accuracy at each reported checkpoint and across all point-level predictions. Its largest advantage occurred during the middle stages of the match, particularly between approximately 40% and 70% match progress, as shown in Figure 2.

Accuracy increased for all models as the match progressed, and the differences between the three plotted models narrowed near the end of the match. However, because accuracy does not assess the quality or confidence of the predicted probabilities, the models were also evaluated using log loss and calibration.

5.2 Log loss + Calibration

To get a better grasp of model performance, the models’ log losses were calculated. Log loss evaluates the quality of the full probability estimates rather than only whether the predicted winner is correct, while penalizing confident incorrect predictions especially strongly. The loss function used for every model was binary log loss, which is defined as

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]. \quad (9)$$

Here, p_i is the predicted probability that player 1 wins, $y_i \in \{0, 1\}$ is the observed match outcome, and N is the total number of predictions. Lower log loss indicates better probability estimates.

Table 5: Model log loss at fixed match-progress checkpoints

Model	25% Loss	50% Loss	75% Loss
Symmetric Markov	0.6292	0.5363	0.3498
Elo-asymmetric Markov	0.5210	0.4549	0.3096
Serve-shrink Markov	0.5142	0.4266	0.2842
HGBM	0.5376	0.3788	0.2299
Markov Ensemble	0.4753	0.3530	0.2002

The Stacked Markov Ensemble performs best across all evaluated match-progress checkpoints. To examine the reliability of its probability estimates more directly, the model was also evaluated using a calibration curve. Calibration curves complement log loss by comparing predicted win probabilities with observed win frequencies.

Following Guo et al. (2017), the predicted probabilities are divided into intervals, or bins, such as (0.60)–(0.75). For each bin B_k , the mean predicted probability is calculated as

$$\text{Pred}_k = \frac{1}{|B_k|} \sum_{i \in B_k} p_i, \quad (10)$$

and the corresponding observed win frequency is calculated as

$$\text{Obs}_k = \frac{1}{|B_k|} \sum_{i \in B_k} y_i, \quad (11)$$

where p_i is the predicted probability that player 1 wins, y_i is the observed match outcome, and $|B_k|$ is the number of predictions in bin B_k . A perfectly calibrated model has an observed win frequency equal to its mean predicted probability in each bin.

Figure 3 plots the mean predicted probability, Pred_k , against the observed win frequency, Obs_k , for each probability interval B_k . Predictions closer to the diagonal line $\text{Obs}_k = \text{Pred}_k$ indicate better calibration.

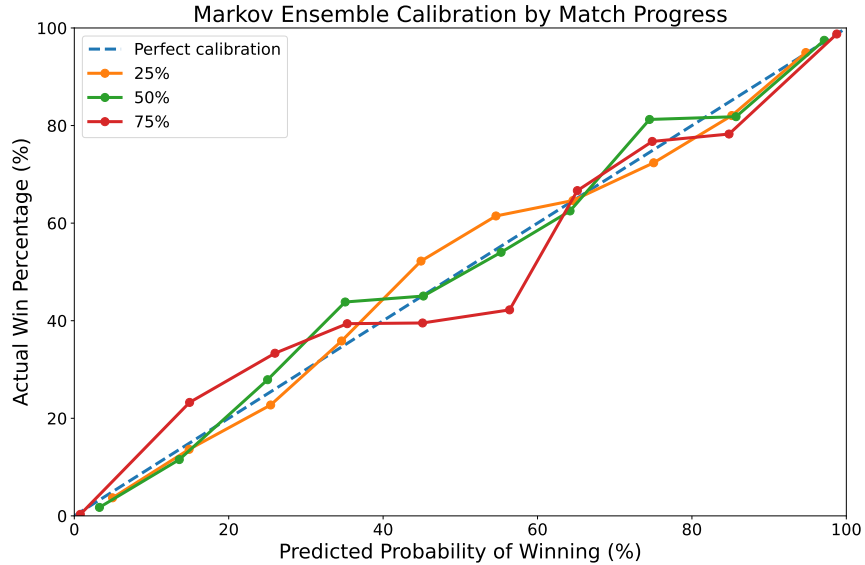


Figure 3: Calibration curve for stacked markov ensemble at 25, 50, and 75% match progress

The reliability diagram indicates that the stacked Markov ensemble is reasonably well calibrated, especially at 25% and 50% match progress. The 75% curve shows larger deviations in the middle probability range, suggesting some late-match overconfidence. Overall, the predicted probabilities generally align with observed win frequencies.

5.3 Tour-specific performance

Figure 4 separates model accuracy by tour for ATP and WTA matches to observe if the two tours differ in serve dynamics, match format, or patterns. Such differences may be obscured when results from both tours are combined.

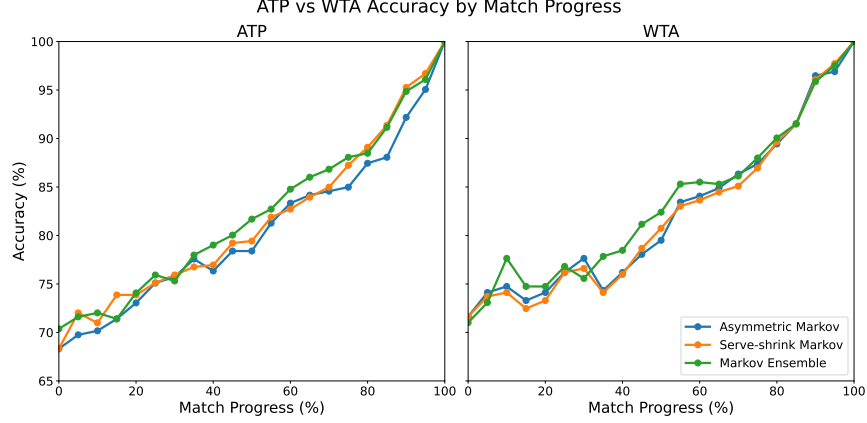


Figure 4: Accuracy of the three models when separated by ATP and WTA

The main pattern is similar on both tours. The stacked Markov ensemble is almost always the strongest model across both ATP and WTA matches, especially through the middle portions of the match. It can also be noticed that the WTA graph fluctuates a lot more than the ATP graph.

6 Discussion

The results indicate that each modeling layer contributes a distinct source of predictive information. The symmetric Markov model confirms that the evolving score state alone contains substantial information about the eventual winner, while the improvement produced by the Elo-asymmetric model demonstrates the value of incorporating pre-match differences in player strength.

The HGBM baseline followed a different performance pattern. Its accuracy was relatively weak early in the match and across all point-level predictions, but it became increasingly competitive as more match information accumulated. At 75% match progress, it achieved the highest accuracy among the individual models and produced lower log loss than all three Markov models at both 50% and 75% progress. These results suggest that direct machine-learning predictions benefit substantially from richer live information, whereas the explicit structure of the Markov models provides a stronger foundation when comparatively little of the match has been observed.

The serve-shrink model produced only small and inconsistent accuracy gains over the Elo-asymmetric model, but achieved lower log loss at every reported checkpoint. This suggests that the live serve updates primarily refined the magnitude of the predicted probabilities rather than frequently changing which player was favored. In many cases, the updated probability remained on the same side of the 0.5 classification threshold, leaving the predicted winner unchanged while producing a probability that better reflected the eventual outcome.

The stacked Markov ensemble achieved the strongest overall performance. Its relative advantage was most apparent during the middle stages of the match, when sufficient live information had accumulated to supplement the pre-match prior but the outcome had not yet become largely determined by the score. By combining the outputs of the structural Markov models with score-state and live-performance features, the HGBM could learn relationships and corrections not fully represented by either Markov model independently. Early in the match, these live features were based on relatively

few observations, whereas late in the match the score state became increasingly decisive and the models’ predictions began to converge. The ensemble’s advantage was therefore smaller near the beginning and end of the match, rather than the model performing poorly during those periods.

The probability-based results further support the value of the ensemble. It achieved the lowest log loss at every reported checkpoint and remained reasonably well calibrated, indicating that its improvement extended beyond winner classification to the quality and reliability of its probability estimates.

Moving on to the results from the tour-specific performance, one possible explanation for the greater WTA local variation is the weaker average server advantage in women’s tennis. Previous research has found that male Grand Slam players serve faster, produce more aces and unreturned serves, and win a greater proportion of service points than female players (Reid et al., 2016). Professional men have also been reported to hold approximately 79% of service games, compared with roughly 66% for women (Goff et al., 2025). More frequent breaks of serve may therefore produce greater variation in the evolving score state and in the predictions of serve-based models, namely the serve-shrink model in this case. This interpretation remains tentative, however, because the observed fluctuations may also reflect sampling variation or differences in match format.

The model accuracies in this paper were also compared with those of DeepTennis (Lerner et al., 2019). DeepTennis used Grand Slam data from 2011–2019, with 2017 as the dev set and 2014 for testing, and reported an overall accuracy of 79.5%. It also reported accuracies of 73.2%, 84.9%, and 89.2% at 25%, 50%, and 75% match progress, respectively.

Because the test year occurs before the development year and within the overall data period, this evaluation is not chronological. If matches occurring after 2014 were included in training, the model may have learned from player form and data patterns that would not have been available when forecasting the 2014 matches. This introduces potential data leakage, or look-ahead bias, and makes the reported results difficult to compare directly with a model evaluated strictly on future seasons.

When the models in this study were evaluated using the same non-chronological split, multiple models matched or exceeded the reported DeepTennis performance:

Table 6: Accuracy under the 2011–2019 non-chronological split

Model	25%	50%	75%	All Points
Symmetric Markov	69.24%	80.64%	87.99%	76.73%
Elo-asymmetric Markov	77.40%	83.05%	88.70%	81.12%
Serve-shrink Markov	77.53%	83.84%	88.83%	81.29%
HGBM	71.81%	81.50%	88.48%	77.53%
Stacked Markov Ensemble	77.66%	83.84%	88.96%	81.30%

It is important to note that this result should not be interpreted as reliable data due to potential data leakage. However, because DeepTennis was the strongest comparable model found, these results do establish Stacked Markov Ensemble as a strong live-prediction tennis model.

Overall, the findings support a hybrid modeling strategy. The Markov recursion represents the scoring rules directly, the serve-shrink update incorporates current serving performance, and the HGBC learns nonlinear relationships and corrections that are difficult to specify manually. The ensemble therefore combines the interpretability of a structured model with the flexibility of machine learning.

7 Limitations

The model represents player strength primarily through serve-point probabilities, although match outcomes also depend on return ability. Future work could estimate separate serve and return strengths and allow these estimates to vary by surface.

Next, other relevant factors are not modeled directly, including fatigue, injury, weather, handedness, playing style, and tactical changes. Although some of these effects may be reflected indirectly in live performance, they are not explicitly represented.

Finally, the stacked Markov ensemble uses a relatively lightweight gradient-boosting model. More advanced models or calibration methods may improve performance, but these gains should be evaluated against the potential loss of interpretability.

8 Conclusion

This paper presents a live tennis win-probability model that combines exact score recursion with learned player-specific inputs. A symmetric Markov model provides a transparent score-only baseline. An Elo-asymmetric model adds pre-match player strength. A serve-shrink model updates the Elo prior using live serve performance. A residual machine-learning model then improves the structural predictions using additional live features.

The results show that each layer adds value. Score structure is useful, pre-match player strength is especially important early, live serve updating becomes more useful as the match progresses, and residual learning provides the strongest overall log-loss performance. The project demonstrates that a compact and interpretable model can produce strong live tennis win-probability estimates by using machine learning where it is most needed: not to learn the rules of tennis, but to estimate and correct the inputs to those rules.

9 Future Work

The main direction for future work is to implement the stacked Markov ensemble as a real-time prediction system. The system would receive point-by-point match data, update the score state and live features after every point, and immediately produce a new match-win probability.

A live interface could display the probability throughout the match, record its progression, and highlight the points that caused the largest changes. Further testing would be needed to ensure that the model remains accurate, calibrated, and computationally efficient when applied to live data streams.

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A Implementation Notes

The project is organized as a small modeling pipeline. The data preparation script builds the point-level table and live running features. The Elo script builds pre-match ratings from ATP and WTA match history. The Markov script contains the point-game-set-match recursion. The model scripts run the symmetric Markov baseline, Elo-asymmetric Markov model, serve-shrink Markov model, HBGM baseline, and Stacked Markov Ensemble model.

The main implementation files are `src/prepare_data.py`, `src/live_features.py`, `src/elo.py`, `src/markov.py`, `src/common.py`, `models/baseline_markov.py`, `models/asymmetric_markov.py`, `models/serve_shrink_model.py`, `models/baseline_gbm.py`, `models/markov_ensemble.py`, and `models/calibration_curve.py`. Code is available at <https://github.com/cx-57/live-tennis-research>.